Watershed Disturbance Mapping System - Technical Specification

Executive Summary

The Watershed Disturbance Mapping System is an automated satellite-based monitoring platform designed to detect and map environmental disturbances across watershed areas. The system leverages multi-temporal remote sensing data, machine learning algorithms, and automated processing workflows to provide early warning and continuous monitoring of habitat-impacting changes.

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1. System Overview and Objectives

Primary Objectives

- **Automated Detection**: Identify watershed disturbances including logging, clearing, fire, flooding, and infrastructure development
- **Early Warning**: Provide timely alerts (69.8% detection within 30 days, 84.6% within 60 days using multi-sensor fusion)
- **Continuous Monitoring**: Establish baseline conditions and track changes over time
- **Habitat Protection**: Support conservation efforts by mapping impacts to sensitive aquatic and riparian habitats

Key Performance Metrics

- **LandTrendr Algorithm**: 86.2% overall accuracy for distinguishing disturbance types
- **Multi-Sensor Fusion (FNRT)**: Median detection lag of 100 days, optimized for early detection
- **Spatial Resolution**: Capable of detecting disturbances down to 0.04 hectares (400 m²) minimum patch size
- **Temporal Coverage**: Monthly processing with historical baseline analysis capability

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2. Data Sources and Satellite Architecture

Primary Data Streams

Landsat Constellation

- Landsat 8 OLI (Operational Land Imager)
- Landsat 9 OLI-2
- 30-meter spatial resolution
- 16-day revisit cycle
- Best for: Historical analysis, consistent long-term monitoring

Sentinel-2 Constellation

- Sentinel-2A and Sentinel-2B
- 10-20 meter spatial resolution
- 5-day revisit cycle (combined constellation)
- Best for: High-frequency monitoring, cloud gap filling

Sentinel-1 SAR (Optional Enhancement)

- C-band Synthetic Aperture Radar
- Cloud-penetrating capability
- 12-day revisit cycle
- Best for: High cloud cover scenarios, all-weather monitoring

Data Access and Storage

- **Platform**: Google Earth Engine (native implementation)
- **Archive Access**: Full Landsat archive (1984-present) and Sentinel-2 archive (2015-present)
- **Processing Environment**: Cloud-based processing with automatic scaling
- **Data Volume**: Approximately 200-500 GB per watershed per year (processed outputs)

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3. Change Detection Algorithms

Algorithm Suite

LandTrendr (Landsat-based Detection and Trending)

Description: Temporal segmentation of pixel trajectories to identify disturbance events and recovery patterns

Technical Specifications:

- Analyzes spectral indices (NDVI, NBR, TCG) across entire Landsat time series
- Segments pixel trajectories into linear trends
- Identifies breakpoints indicating disturbance events
- Distinguishes between disturbance types based on spectral-temporal signatures

Strengths:

- 86.2% accuracy in distinguishing disturbance types
- Excellent for historical analysis
- Captures gradual and abrupt changes
- Native Google Earth Engine implementation

Optimal Use Cases:

- Long-term trend analysis
- Disturbance type classification (fire, harvest, gradual degradation)
- Recovery trajectory assessment

Multi-Sensor Fusion (FNRT - Forest Near Real-Time)

Description: Combines Landsat 8/9 and Sentinel-2 data streams with optional Sentinel-1 integration for rapid detection

Technical Specifications:

- Fuses optical data from multiple sensors
- Normalized spectral indices across platforms
- Statistical outlier detection on harmonized time series
- Confidence scoring based on observation frequency and magnitude of change

Performance Metrics:

- Early Detection: 69.8% within 30 days
- High Detection: 84.6% within 60 days
- Median Detection Lag: 100 days
- False Positive Rate: <15% with confidence threshold >0.8

Optimal Use Cases:

- Near real-time monitoring
- Early warning systems
- High cloud cover environments (when combined with SAR)
- Rapid response requirements

Spectral Indices Calculated

NDVI (Normalized Difference Vegetation Index)

- Formula: (NIR - Red) / (NIR + Red)
- Primary indicator of vegetation health and density
- Range: -1 to +1 (vegetation typically 0.2 to 0.8)

NBR (Normalized Burn Ratio)

- Formula: (NIR - SWIR2) / (NIR + SWIR2)
- Optimized for fire and burn scar detection
- Sensitive to vegetation moisture content

TCG (Tasseled Cap Greenness)

- Linear transformation of multiple spectral bands
- Emphasizes green vegetation signal
- Effective for forest disturbance detection

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4. Automated Processing Pipeline

Monthly Workflow Schedule

Execution Frequency: Every 5 days (Sentinel-2 revisit schedule)

Processing Steps

Step 1: Imagery Acquisition

- **Schedule**: Automated query every 5 days
- **Data Retrieved**:
  - Sentinel-2 L2A (atmospherically corrected surface reflectance)
  - Landsat 8/9 Collection 2 Level-2
  - Cloud cover metadata
- **Spatial Extent**: Defined watershed boundary + 5 km buffer
- **Temporal Window**: Last 90 days

Step 2: Atmospheric Correction (Sen2Cor)

- **Algorithm**: Sen2Cor for Sentinel-2, USGS standard for Landsat

- **Outputs**: Bottom-of-atmosphere reflectance
- **Correction Factors**:
  - Aerosol optical depth
  - Water vapor content
  - Ozone concentration
  - Terrain illumination effects

Step 3: Cloud Masking

- **Method**: s2cloudless algorithm for Sentinel-2, CFMask for Landsat
- **Cloud Probability Threshold**: 30% (pixels above threshold rejected)
- **Cloud Shadow Detection**: Geometric projection based on sun angle
- **Quality Filtering**: Remove pixels with saturated or missing values
- **Minimum Valid Observations**: 3 per 90-day period required for analysis

Step 4: Spectral Index Calculation

- **Indices Computed**: NDVI, NBR, TCG for each acquisition
- **Normalization**: Z-score standardization relative to historical baseline
- **Composite Generation**: Maximum NBR, median NDVI per period

Step 5: Inference (U-Net Deep Learning Model)

- **Model Architecture**: U-Net convolutional neural network
- **Input Features**:
  - Multi-temporal spectral indices (t, t-1, t-2)
  - Texture metrics (GLCM)
  - Topographic variables (slope, aspect, elevation)
- **Output**: Pixel-wise disturbance probability (0-1 scale)
- **Training Data**: 50,000+ labeled disturbance polygons across Pacific Northwest watersheds
- **Model Performance**: 89% precision, 84% recall on validation set

Step 6: Change Detection (Baseline Comparison)

- **Baseline**: Rolling 3-year median of spectral indices (updated annually)
- **Change Threshold**: Deviation >2.5 standard deviations from baseline
- **Temporal Logic**: Change must persist across 2+ observations to be flagged
- **Magnitude Assessment**: Calculate change intensity (mild, moderate, severe)

Step 7: Validation and Confidence Scoring

- **Confidence Score Calculation**:
  - U-Net probability (40% weight)
  - Number of confirming observations (30% weight)
  - Magnitude of spectral change (20% weight)
  - Spatial pattern consistency (10% weight)
- **Filtering Threshold**: Confidence >0.8 for alert generation
- **Spatial Coherence**: Apply morphological filtering to remove isolated pixels

Step 8: PostGIS Database Update

- **Database Schema**:
  - `detections` table: Geometry, detection_date, confidence_score, disturbance_type, area_hectares

    - `time_series` table: Historical spectral values per pixel
    - `baselines` table: Reference conditions by ecosystem type
- **Indexing**: Spatial (GiST) and temporal (B-tree) indices for rapid queries
- **Versioning**: Maintain detection history with status flags (new, confirmed, false_positive, resolved)

Step 9: Alert Generation

- **Alert Triggers**:
    - New high-confidence detection (confidence >0.8)
    - Detection area >0.5 hectares
    - Proximity to sensitive habitats (<100m from streams, within riparian zones)
- **Notification Methods**:
    - Email to registered stakeholders
    - SMS via Twilio for critical alerts
    - API webhook to external systems
- **Alert Content**:
    - Location (coordinates, map thumbnail)
    - Detection date and confidence
    - Estimated disturbance area
    - Spectral change characteristics
    - Link to web dashboard for details

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5. Quality Control Framework

Data Quality Thresholds

Cloud Coverage Filtering

- **Threshold**: 30% cloud probability per pixel
- **Rationale**: Balance between data availability and accuracy
- **Impact**: Reduces false positives from cloud shadow and edge artifacts

Minimum Clear Observations

- **Requirement**: 3 valid observations per 90-day window
- **Rationale**: Ensures sufficient temporal sampling for change detection
- **Handling**: Flag regions with insufficient data; supplement with SAR when available

Confidence Score Filtering

- **Alert Threshold**: >0.8 confidence for automated alerts
- **Review Threshold**: 0.6-0.8 for manual review queue
- **Rejection Threshold**: <0.6 automatically filtered out
- **Calibration**: Thresholds tuned to maintain <10% false positive rate

Spatial Filtering

- **Minimum Patch Size**: 0.04 hectares (4 pixels at 10m resolution)
- **Rationale**: Eliminates spurious single-pixel detections and sensor noise
- **Method**: Morphological opening operation with 2-pixel kernel

Temporal Persistence

- **Requirement**: Disturbance must appear in 2+ consecutive observations

- **Temporal Window**: Within 30-day period
- **Exception**: Very high confidence detections (>0.95) can trigger on single observation

Validation Protocols

Automated Cross-Validation

- Compare detections across LandTrendr and FNRT algorithms
- Flag discrepancies for manual review
- Agreement between methods increases confidence score by 0.1

Field Validation Program

- Annual field surveys of random sample (5% of detections)
- Ground truth data collection: photos, GPS points, disturbance characterization
- Accuracy assessment and model recalibration

User Feedback Loop

- Dashboard allows users to flag false positives/negatives
- Feedback integrated into training data for model updates
- Quarterly model retraining incorporating validated corrections

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6. System Architecture and Infrastructure

Technology Stack

Processing Platform: Google Earth Engine

- JavaScript/Python API for algorithm development
- Distributed computing for large-scale analysis
- Built-in data catalog access

Web Application: Python/Django Backend

- RESTful API for data access
- User authentication and role-based access control
- Task scheduling with Celery

Database: PostgreSQL 14 + PostGIS 3.2

- Stores detection results, time series data, user accounts
- Spatial indexing for rapid geometric queries
- Partitioned tables for time-series optimization

Frontend: React.js + Leaflet

- Interactive map interface for exploring detections
- Time-slider for temporal analysis
- Downloadable reports and data exports

Alert System:

- Twilio API for SMS notifications
- SendGrid for email delivery

- Webhook support for third-party integrations

Deployment Architecture

Cloud Infrastructure: AWS

- EC2 instances for web servers (auto-scaling group)
- RDS PostgreSQL for database (Multi-AZ deployment)
- S3 for imagery storage and backups
- Lambda functions for lightweight processing tasks
- CloudWatch for monitoring and logging

Scalability Considerations

- Horizontal scaling of web servers based on load
- Database read replicas for reporting queries
- Caching layer (Redis) for frequently accessed data
- CDN (CloudFront) for static assets

Security Measures

- HTTPS/TLS 1.3 for all communications
- OAuth 2.0 for API authentication
- Role-based access control (admin, analyst, viewer)
- Database encryption at rest (AES-256)
- Regular security audits and penetration testing

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7. User Interface and Visualization

Web Dashboard Components

Map View

- Base layers: Satellite imagery, topographic maps, aerial photos
- Overlays: Detections (color-coded by confidence), watershed boundaries, stream networks, protected areas
- Temporal controls: Date range selector, animation player for time-lapse
- Interactive features: Click detection for details, draw tools for custom areas of interest

Detection Details Panel

- Geometry and area
- Detection date and confidence score
- Spectral change graphs (NDVI, NBR time series)
- Before/after imagery comparison
- Disturbance type classification
- Export options (GeoJSON, Shapefile, KML)

Analytics Dashboard

- Summary statistics: Total detections, area disturbed, detection rate over time
- Charts: Disturbance types by area, monthly detection trends, confidence distribution
- Reporting: Automated monthly/quarterly reports (PDF generation)

Alert Management

- Configure alert preferences (email, SMS, webhook)
- Filter alerts by confidence, area, proximity to features
- Alert history and status tracking
- Mute/unmute specific geographic areas

Mobile Application (Optional Enhancement)

- Offline-capable field data collection
- Camera integration for ground truth photos
- GPS tracking for field validation surveys
- Syncs with main database when connectivity available

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8. Limitations and Failure Modes

Technical Limitations

1. Submerged Vegetation (>3m depth)

Issue: Optical satellites cannot penetrate deep water

Impact:

- Aquatic plants below 3m depth are undetectable
- Changes in submerged aquatic vegetation (SAV) beds may go unnoticed

Affected Species/Habitats:

- Deep-water macrophyte communities
- Bottom-dwelling aquatic plants in turbid water

Workaround:

- Use sonar/acoustic surveys for deep-water habitats
- Focus optical monitoring on shallow (<2m) littoral zones

2. Turbid Water Environments

Issue: High sediment or algal content reduces water clarity

Impact:

- 50% accuracy reduction in turbid conditions
- Cannot distinguish submerged features
- Increased false negatives in muddy rivers/estuaries

Affected Environments:

- Glacial-fed rivers
- Estuaries with high sediment load
- Post-storm runoff conditions

Mitigation:

- Prioritize clear-water seasons for assessment

- Use water quality data to filter unreliable detections
- Combine with in-situ monitoring in chronically turbid systems

3. Small Patch Detection Threshold

Issue: Sentinel-2 10m resolution limits minimum detectable size

Impact:

- Patches <100m² (0.01 hectares) below reliable detection threshold
- Linear features <10m wide may be missed (narrow roads, small streams)

Affected Features:

- Small forest gaps
- Narrow riparian clearing
- Individual tree mortality

Mitigation:

- Use high-resolution imagery (Planet Labs, drones) for fine-scale assessment
- Field surveys for small-scale change validation
- Focus on ecologically significant (>0.5 ha) disturbances

4. Cloud Cover Interference

Issue: Persistent cloud cover creates observation gaps

Impact:

- Multi-week gaps possible in cloudy regions (Pacific Northwest)
- Delayed detection in high-cloud seasons (winter)
- Reduced temporal resolution of time-series

Affected Regions:

- Temperate rainforest zones
- Coastal watersheds
- Winter season (November-March)

Mitigation:

- Multi-temporal analysis to fill gaps
- Sentinel-1 SAR integration for cloud-free observations
- Extend detection window during cloudy periods
- Prioritize summer/fall imagery for annual assessments

5. Species Confusion (Plant Identification)

Issue: Optical spectral signatures may overlap between species

Impact:

- 10-20% confusion rate with spectrally similar native species
- Cannot reliably distinguish between closely related species
- May mistake native expansion for invasive spread

Affected Applications:

- Invasive species-specific monitoring (e.g., distinguishing native vs. invasive knotweed)
- Rare plant population tracking

Mitigation:

- Combine with field surveys for species-level identification
- Use phenology (seasonal timing) to differentiate species
- Hyperspectral imagery (if available) for finer spectral discrimination
- Focus on genus-level or community-level changes

Operational Failure Modes

Processing Failures

- Google Earth Engine service disruptions
- API rate limit exhaustion
- Memory errors on large watershed processing

Data Availability Issues

- Satellite downtime (rare)
- Data distribution delays
- Missing or corrupted imagery

Database Connectivity

- Network outages
- Database maintenance windows
- Replication lag

Alert Delivery Failures

- Email/SMS service disruptions
- Invalid contact information
- Webhook endpoint failures

Mitigation Strategies:

- Automated retry logic with exponential backoff
- Queue-based processing for resilience
- Health monitoring and automated alerts for system issues
- Redundant alert channels (email + SMS + dashboard)

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9. Mitigation Strategies and System Enhancement

Addressing Core Limitations

Multi-Sensor Integration

Strategy: Combine optical with radar and lidar data sources

Implementation:

- Integrate Sentinel-1 SAR for cloud penetration
- Use Planet Labs high-resolution tasking for small-patch validation
- Incorporate drone surveys for sub-meter resolution in priority areas

Benefits:

- Reduces cloud cover impact by 60-70%
- Extends detection capability to smaller features
- Provides all-weather monitoring capability

Field Validation Program

Strategy: Systematic ground-truthing of detections

Implementation:

- Random stratified sampling: 5% of detections per quarter
- Targeted surveys of low-confidence detections (0.6-0.8 range)
- Citizen science engagement for opportunistic validation
- Photo point monitoring at fixed stations

Benefits:

- Validates model performance
- Identifies systematic errors for correction
- Builds training dataset for model improvement
- Engages community stakeholders

Multi-Temporal Analysis for Cloud Gaps

Strategy: Leverage extended time series to fill observation gaps

Implementation:

- Expand analysis window to 180 days in high-cloud regions
- Use temporal interpolation for missing observations
- Weight recent observations higher than older ones
- Flag low-confidence detections in data-sparse periods

Benefits:

- Maintains detection capability during cloudy seasons
- Reduces seasonal bias in detection rates
- Provides more robust change trajectories

High-Resolution Drone Surveys

Strategy: Targeted UAS flights for detailed mapping of priority sites

Implementation:

- Deploy drones for post-detection detailed assessment
- Create high-resolution orthomosaics (2-5cm resolution)
- Generate digital surface models for vegetation structure
- Conduct seasonal repeat flights for change tracking

Benefits:

- Confirms satellite detections with cm-scale imagery
- Enables species-level identification
- Detects sub-meter changes invisible to satellites
- Provides detailed baseline for restoration monitoring

Planet Labs High-Priority Site Monitoring

Strategy: Supplement Sentinel/Landsat with daily 3m resolution imagery

Implementation:

- Subscribe to PlanetScope imagery for critical watersheds
- Daily revisit for near-real-time monitoring
- Automated change detection at 3m resolution
- Integration with main processing pipeline

Benefits:

- Daily observation frequency (vs. 5-day Sentinel-2)
- 3m spatial resolution (vs. 10m Sentinel-2)
- Reduces detection lag to <7 days
- Fills cloud gaps more effectively

Model Improvement Roadmap

Year 1: Baseline Enhancement

- Expand training dataset to 100,000+ labeled samples
- Incorporate regional model variants (coastal, alpine, forested)
- Implement automated active learning for difficult cases

Year 2: Advanced Features

- Integrate SAR (Sentinel-1) for all-weather detection
- Add lidar-derived topographic and vegetation structure features
- Develop disturbance type classifier (fire, harvest, flood, urban expansion)

Year 3: Real-Time Capabilities

- Reduce median detection lag to <30 days
- Implement continuous monitoring (vs. monthly batch)
- Develop predictive models for disturbance risk

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10. Deployment and Operational Guidelines

Initial System Setup

Phase 1: Watershed Boundary Definition (Week 1-2)

- Delineate watershed boundaries using DEM and hydrologic modeling
- Define areas of interest (stream buffers, riparian zones, sensitive habitats)
- Establish processing extent with 5 km buffer for edge effects

Phase 2: Baseline Data Generation (Week 3-8)

- Collect 3-year historical Landsat/Sentinel-2 archive
- Generate baseline spectral indices and statistical summaries
- Compute reference conditions by land cover type and elevation zone
- Create cloud-free composite imagery for baseline visualization

Phase 3: Model Training and Calibration (Week 9-12)

- Collect training data: digitize 500+ disturbance polygons per watershed
- Train or fine-tune U-Net model for local conditions
- Validate model performance: target >85% precision, >80% recall
- Optimize confidence thresholds based on user tolerance for false positives

Phase 4: Processing Pipeline Deployment (Week 13-14)

- Configure automated scheduling (every 5 days)
- Set up database schema and initial data loading
- Deploy web dashboard and alert system
- Conduct end-to-end system testing

Phase 5: User Training and Launch (Week 15-16)

- Train staff on dashboard use, alert interpretation, and data export
- Establish protocols for field validation and feedback
- Launch system with trial period (manual review of all alerts)
- Collect user feedback and refine alert thresholds

Ongoing Operations

Monthly Tasks

- Review processing logs for failures or anomalies
- Validate high-confidence detections with aerial imagery or field visits
- Update disturbance database with confirmed changes
- Generate monthly summary reports for stakeholders

Quarterly Tasks

- Analyze detection statistics and model performance
- Conduct field validation surveys (5% sample)
- Update training dataset with new confirmed detections
- Retrain models if significant drift detected

Annual Tasks

- Update baseline conditions (rolling 3-year median)
- Comprehensive accuracy assessment report
- System performance optimization (query tuning, archiving old data)
- Strategic planning for new features or expanded coverage

Maintenance and Support

System Monitoring

- Automated health checks every 15 minutes
- Alert on processing failures (email to admin)
- Weekly performance reports (processing time, data volume, detection counts)

Software Updates

- Quarterly security patches
- Biannual feature releases
- Google Earth Engine API updates as released
- Database maintenance windows (monthly, off-hours)

User Support

- Email support (48-hour response time)
- Online documentation and tutorial videos
- Quarterly user webinars for new features and best practices
- Annual user survey for feedback and prioritization

Cost Considerations

Initial Setup (One-Time)

- Cloud infrastructure setup: $5,000
- Training data collection and model development: $20,000
- Dashboard and database development: $30,000
- **Total Initial Investment: ~$55,000**

Annual Operating Costs

- Cloud hosting (AWS): $12,000/year
- Google Earth Engine (if commercial use): $15,000/year
- Satellite data subscriptions (if Planet Labs added): $25,000/year
- Staff time (2 FTE: analyst + developer): $150,000/year
- Field validation program: $10,000/year
- **Total Annual Operating Cost: ~$212,000/year (without Planet Labs: ~$187,000)**

Success Metrics

Technical Performance

- Detection Precision: >85%
- Detection Recall: >80%
- Median Detection Lag: <100 days
- False Positive Rate: <15%

Operational Efficiency

- System Uptime: >99%
- Processing Time: <24 hours per batch
- Alert Delivery: <1 hour from detection confirmation

User Satisfaction

- Dashboard User Adoption: >80% of staff
- Alert Actionability: >70% of alerts lead to field verification or management action
- User Satisfaction Score: >4/5

Future Enhancements

Short-Term (Year 1-2)

- Mobile app for field data collection
- Automated change type classification (fire, harvest, flood, development)
- Integration with additional satellite sources (Planet, Maxar)

Medium-Term (Year 2-3)

- Predictive modeling: identify areas at high risk of disturbance
- Watershed health dashboard: integrate with water quality, flow, and habitat data
- Automated reporting for regulatory compliance

Long-Term (Year 3+)

- Machine learning for restoration success monitoring
- Integration with climate models for disturbance forecasting
- Expansion to real-time monitoring (<7 day detection lag)
- Development of open-source community version

Conclusion

The Watershed Disturbance Mapping System provides a scalable, automated solution for monitoring environmental changes across large watershed areas. By integrating multiple satellite data sources, advanced change detection algorithms, and robust quality control measures, the system delivers timely and accurate disturbance alerts to support conservation and management decisions. While inherent limitations exist related to spatial resolution, cloud cover, and species-level discrimination, strategic mitigation approaches including multi-sensor integration, field validation, and targeted high-resolution surveys can significantly enhance system performance. With proper implementation and ongoing refinement, this system serves as a powerful tool for protecting critical aquatic and riparian habitats.

Document Version: 1.0
Last Updated: October 29, 2025
Status: Draft Technical Specification